

IoT Practical: Oscilloscope

CMD commands:

```
sudo raspi-config (enable i2c)
```

```
pip install board --break-system-packages
```

```
sudo pip install drawnow --break-system-packages
```

```
sudo apt-get install -y i2c-tools python3-smbus
```

```
python3 -m pip install --upgrade --no-cache-dir adafruit-blinka adafruit-circuitpython-busdevice  
adafruit-circuitpython-ads1x15 --break-system-packages
```

Connection:

ADS1115 Pin	Raspberry Pi Pin Number	Raspberry Pi Function
VDD	Pin 1	3.3V Power
GND	Pin 6	Ground
SDA	Pin 3	GPIO2 (SDA1 – I ² C Data)
SCL	Pin 5	GPIO3 (SCL1 – I ² C Clock)

Code:

```
from collections import deque  
  
import time  
  
import board  
  
import busio  
  
import numpy as np  
  
import matplotlib.pyplot as plt  
  
from adafruit_ads1x15.ads1115 import ADS1115  
from adafruit_ads1x15.analog_in import AnalogIn  
from matplotlib.animation import FuncAnimation  
  
# ===== SETTINGS =====  
  
DURATION = 10          # seconds shown on screen  
  
SAMPLE_RATE = 128      # ADS1115: 8–860 SPS  
  
GAIN = 1                # ±4.096V
```

```

CHANNEL = 0          # A0

I2C_ADDRESS = 0x48

# =====

# I2C setup

i2c = busio.I2C(board.SCL, board.SDA)

# ADC setup

ads = ADS1115(i2c, address=I2C_ADDRESS)

ads.data_rate = SAMPLE_RATE

ads.gain = GAIN

chan = AnalogIn(ads, CHANNEL)

# Buffer

num_samples = SAMPLE_RATE * DURATION

x = np.linspace(0, DURATION, num_samples)

y = deque([0.0] * num_samples, maxlen=num_samples)

# Plot

fig, ax = plt.subplots()

ax.set_xlim(0, DURATION)

ax.set_ylim(-4.2, 4.2)

ax.set_xlabel("Time [s]")

ax.set_ylabel("Voltage [V]")

ax.set_title("ADS1115 Software Oscilloscope")

ax.grid(True)

line, = ax.plot(x, y, lw=1)

# Update loop

def update(frame):

```

```
try:
    voltage = chan.voltage
except Exception:
    voltage = 0.0
y.append(voltage)
line.set_ydata(y)
return line,
```

```
# Animation
```

```
ani = FuncAnimation(
    fig,
    update,
    interval=1000 / SAMPLE_RATE,
    blit=True
)
```

```
plt.show()
```